

High-Order Explicit Heat Equation Solver Verification Report for `example5`

Generated report for `testHighOrderHeatEquationExplicit`

May 7, 2026

Abstract

This report updates the verification documentation for `testHighOrderHeatEquationExplicit` using the current `example5` results in `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationExplicit/results/example5`. Its structure follows the report used for `highOrderManufacturedFDAExplicit`, but the model is simpler: only the scalar heat variable T is advanced, and there are no ionic ODE states or local state reconstructions. The report compares the standard second-order finite-volume operator (`hoT_N0`) with LRE high-order reconstructions for T .

1 Governing Equation

The solver advances a transient scalar diffusion problem,

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (\alpha \nabla T) + Q. \quad (1)$$

The explicit cell update is

$$T_K^{n+1} = T_K^n + \frac{\Delta t}{\rho C_p} R_K^n. \quad (2)$$

For `example5`, the implementation uses $R_K^n = L_h(T^n)_K - Q_K^n$, where Q is the exact FDA-like manufactured ionic source reused as a scalar heat-equation source.

2 Manufactured Example5

The exact temperature is

$$T^*(\mathbf{x}, t) = A\sqrt{1+t}F(\mathbf{x}). \quad (3)$$

For two-dimensional runs,

$$F(x, y) = \cos(\pi x) \cos(2\pi y), \quad G(x, y) = 1 + xy^2. \quad (4)$$

The source is built from

$$Q^*(\mathbf{x}, t) = AI_{\text{ion}}^*(\mathbf{x}, t), \quad (5)$$

where

$$I_{\text{ion}}^* = -\frac{1}{2} (u_1^* + u_3^* - T^*) (u_2^*)^2 (T^* - u_3^*) + \beta_T (T^* - u_3^*), \quad (6)$$

$$u_1^* = (1+t)G + T^*, \quad u_2^* = \frac{1}{(1+t)\sqrt{G}}, \quad u_3^* = 0. \quad (7)$$

These u_i^* are analytical helper functions only; unlike the FDA solver, this heat solver does not solve ODEs.

3 Spatial Discretization

When `useHighOrder_T=false`, diffusion is computed with the standard OpenFOAM finite-volume Laplacian. When `useHighOrder_T=true`, face fluxes are computed using LRE-reconstructed gradients,

$$\Phi_f \approx |S_f| \sum_q \omega_{fq} \mathbf{n}_f \cdot \alpha \nabla_h T(\mathbf{x}_{fq}), \quad (8)$$

and assembled from `fvc::div(fluxT_H0)`. The labels `p1`, `p2`, and `p3` denote first-, second-, and third-order Taylor terms. The label `N0` denotes the standard second-order finite-volume calculation.

4 Boundary Conditions

For the high-order flux, `zeroGradient` and `empty` patches are treated as no-flux boundaries by setting the high-order boundary flux to zero,

$$\int_{\partial K \cap \Gamma_N} \mathbf{n} \cdot \alpha \nabla T \, dS = 0. \quad (9)$$

Fixed-value patches keep their boundary contribution in the high-order reconstruction.

5 Time Integration

Time is advanced with forward Euler. There is no implicit linear system, no ODE solve, and no consistent mass matrix in this solver; the finite-volume update corresponds to a diagonal cell-volume mass. Large time steps can therefore produce divergent spatial fits, which appear as large negative fitted orders in the tables.

6 Error Measures and Rates

The solver writes cell-centred errors such as

$$L_2 = \left(\frac{\sum_K |T_K - T_K^*|^2 |K|}{\sum_K |K|} \right)^{1/2} \quad (10)$$

and reconstructed quadrature errors G_2 , computed by reconstructing T_h at cell quadrature points and comparing with T^* there. The expected spatial orders are approximately 2 for `N0` and `p1`, 3 for `p2`, and 4 for `p3`.

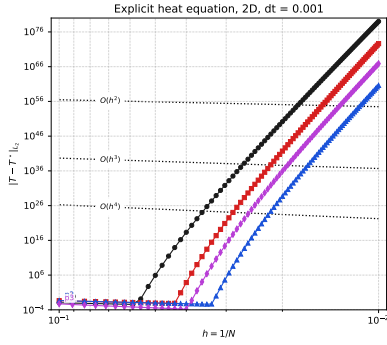
7 2D Results

The available 2D data are read from `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationExplicit/results/example5/2D`. Table 1 reports the all-mesh log-log spatial fits stored in the `spatial_convergence_rate_dt_*_total.dat` files.

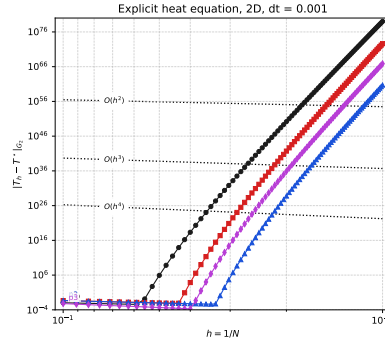
7.1 2D Convergence Figures

Table 1: All-mesh fitted spatial convergence orders for the explicit heat solver, 2D example5 results. Expected orders are 2 for NO/p1, 3 for p2, and 4 for p3.

HO T	Δt	Expected	p_{L_2}	p_{G_2}
NO	10^{-3}	2	-96.359	-96.359
NO	10^{-4}	2	-97.205	-97.205
NO	10^{-5}	2	2.002	2.002
NO	10^{-6}	2	2.002	2.002
p1	10^{-3}	2	-94.180	-94.146
p1	10^{-4}	2	-82.529	-82.504
p1	10^{-5}	2	2.224	2.200
p1	10^{-6}	2	2.224	2.200
p2	10^{-3}	3	-81.411	-81.521
p2	10^{-4}	3	-10.256	-10.281
p2	10^{-5}	3	2.080	2.088
p2	10^{-6}	3	2.080	2.088
p3	10^{-3}	4	-89.839	-89.690
p3	10^{-4}	4	-77.590	-77.464
p3	10^{-5}	4	4.059	4.173
p3	10^{-6}	4	4.054	4.169

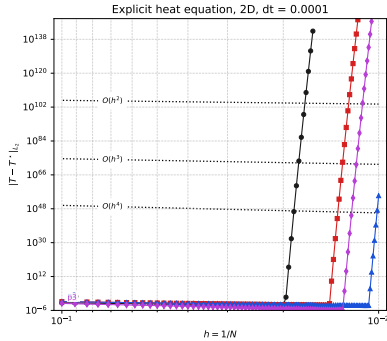


(a) $\|T - T^*\|_{L_2}$

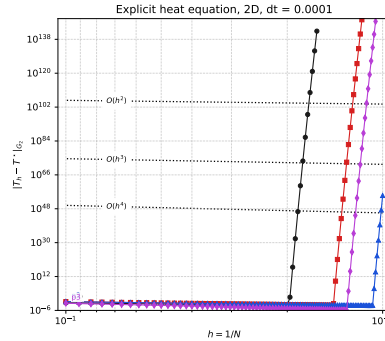


(b) $\|T_h - T^*\|_{G_2}$

Figure 1: Explicit high-order heat-equation solver, 2D example5 results for $\Delta t = 10^{-3}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

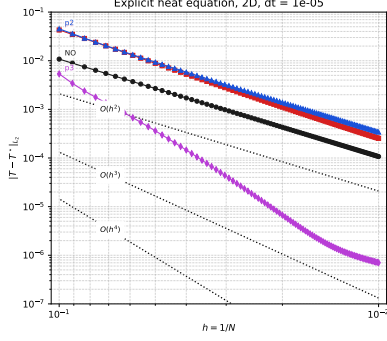


(a) $\|T - T^*\|_{L_2}$

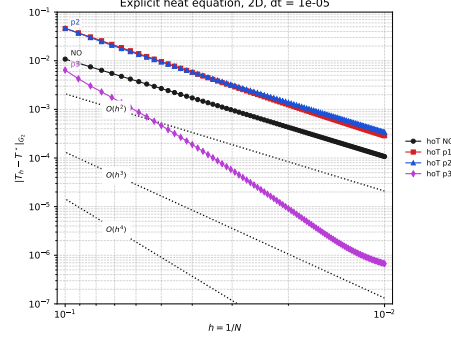


(b) $\|T_h - T^*\|_{G_2}$

Figure 2: Explicit high-order heat-equation solver, 2D example5 results for $\Delta t = 10^{-4}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

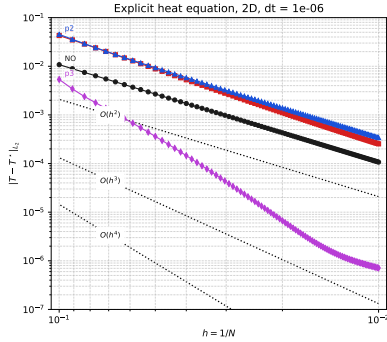


(a) $\|T - T^*\|_{L_2}$

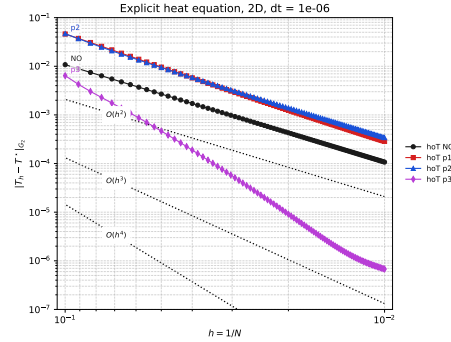


(b) $\|T_h - T^*\|_{G_2}$

Figure 3: Explicit high-order heat-equation solver, 2D example5 results for $\Delta t = 10^{-5}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.



(a) $\|T - T^*\|_{L_2}$



(b) $\|T_h - T^*\|_{G_2}$

Figure 4: Explicit high-order heat-equation solver, 2D example5 results for $\Delta t = 10^{-6}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

8 Three-Dimensional Results

No `example5/3D` result directory was present when this report was generated. The report and plotting script are organized so that three-dimensional data can be added later with the same folder structure.

9 Discussion

The current `example5` dataset contains only two-dimensional runs. The smaller time steps show the intended second-order behaviour for the standard operator and approximately fourth-order behaviour for `p3` in L_2 and G_2 . The larger time steps, especially 10^{-3} and in several cases 10^{-4} , are dominated by explicit time-step instability or temporal error, so their fitted spatial rates should not be interpreted as asymptotic spatial convergence.

10 Files

- Solver source: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationExplicit.C`
- Tutorial results: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationExplicit/results/example5`
- Plot script: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationExplicit/report/plot_convergence.m`